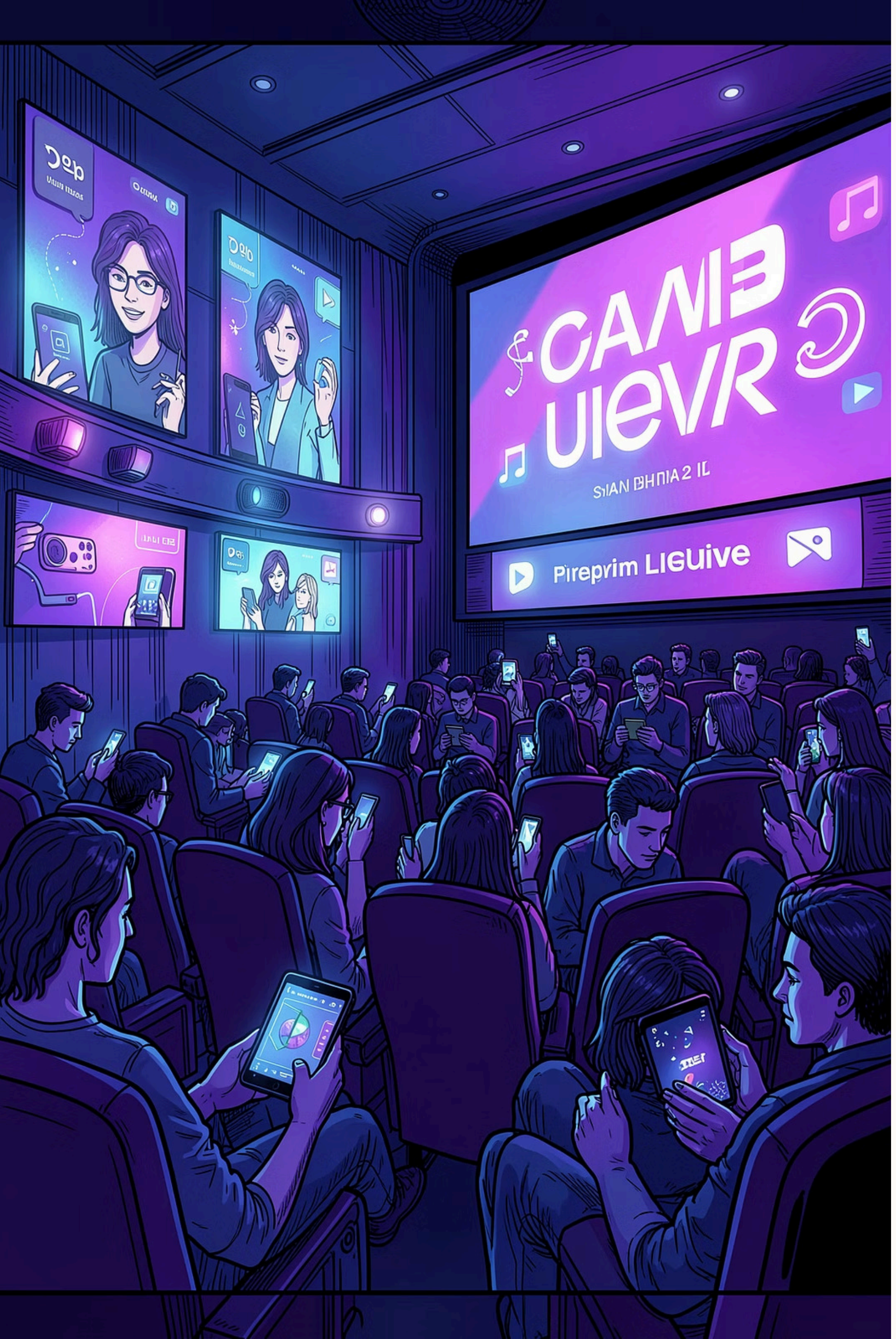

Netflix Movie Recommendation System using Collaborative Filtering and SVD Matrix Factorization

Personalized content discovery using recommendation systems

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Problem Statement

Challenge

Streaming platforms contain thousands of movies; users struggle to discover relevant content.

Goal

Build a recommender that predicts preferences and suggests unseen movies.

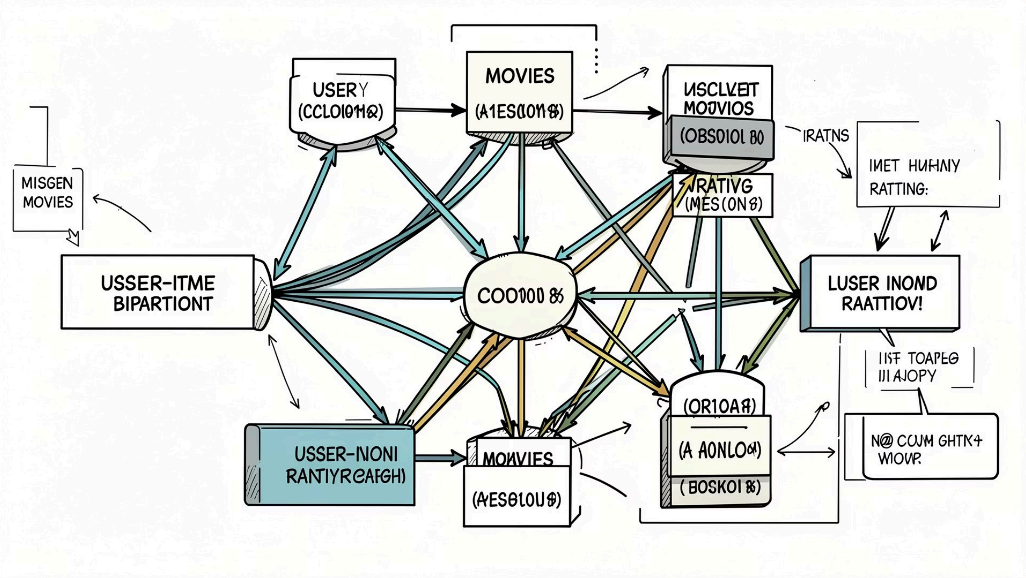
Objectives

Analyse ratings, develop models, compare performance and generate personalised recommendations.

Dataset Overview — Netflix Prize Dataset

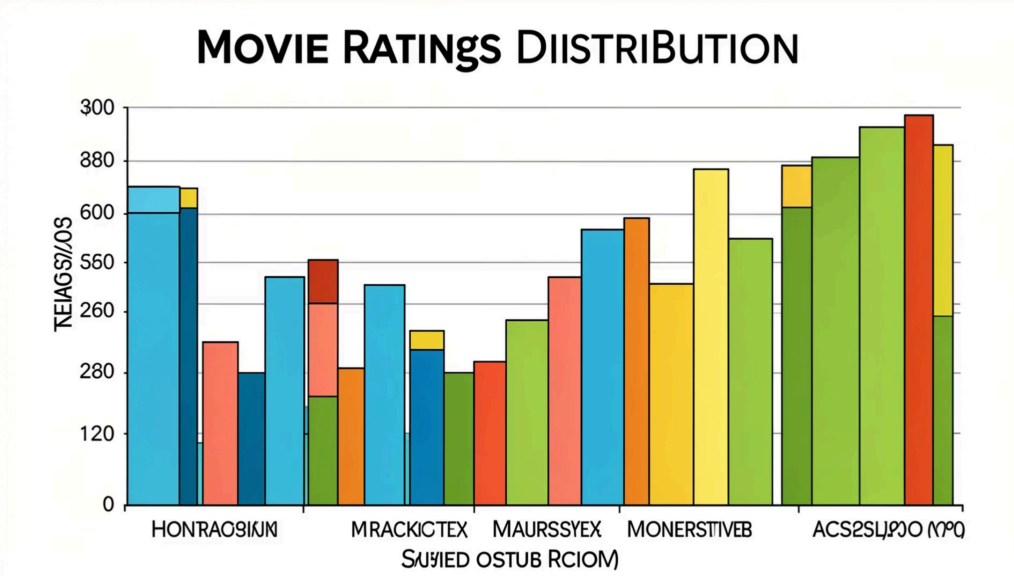
Metric	Value	Notes
Ratings	50,000	Sample size
Users	24,430	Unique users
Movies	504	Unique titles

Key challenge: highly sparse user–item interaction matrix.

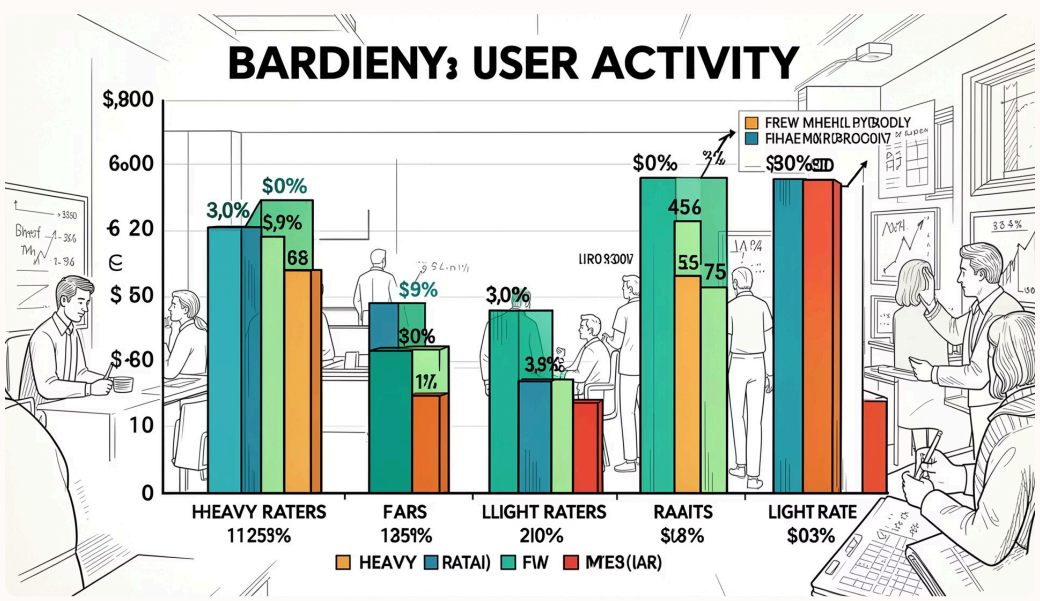


Visual: user–movie–rating relationships showing many missing interactions.

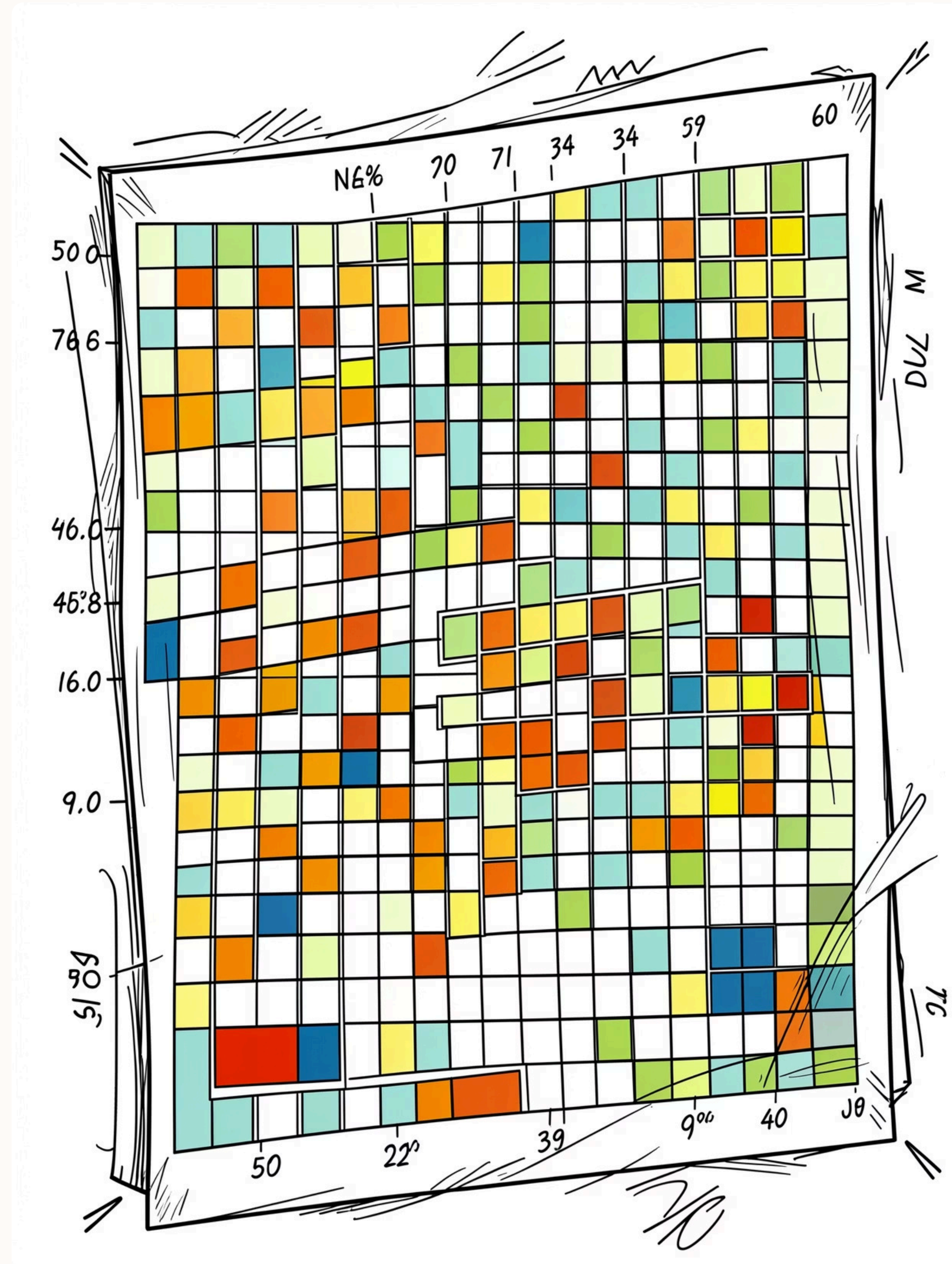
Exploratory Data Analysis



Rating distribution: skewed toward positive values.



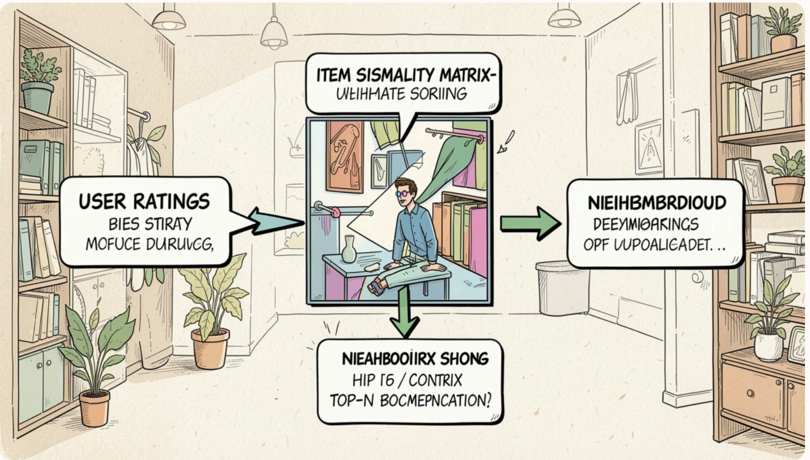
User activity: uneven engagement across users.



Dataset sparsity: majority of user–movie pairs unobserved.

These findings motivate methods that handle sparse interactions.

Model 1 — Item-Based Collaborative Filtering



Approach

Use similarity between movies to recommend items based on user interactions with similar movies.

Result

RMSE = 1.2377

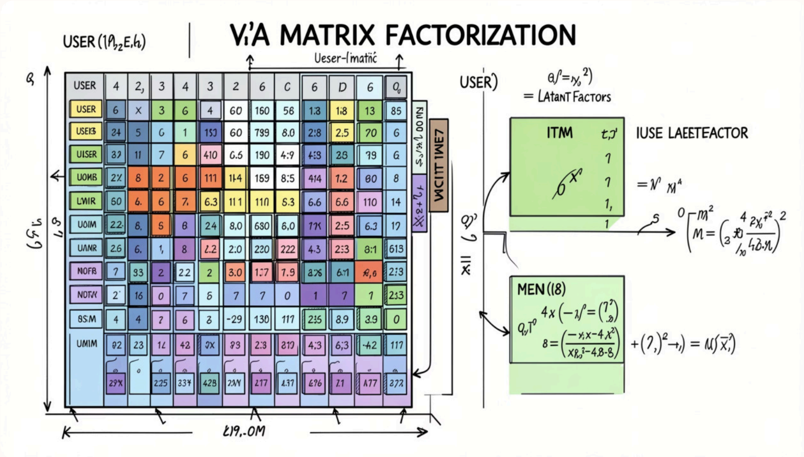
Pros

Easy to interpret; widely used.

Cons

Sensitive to sparsity; performance degrades with few co-ratings.

Model 2 — SVD Matrix Factorization



Approach

Learn latent user and item factors to capture hidden preferences.

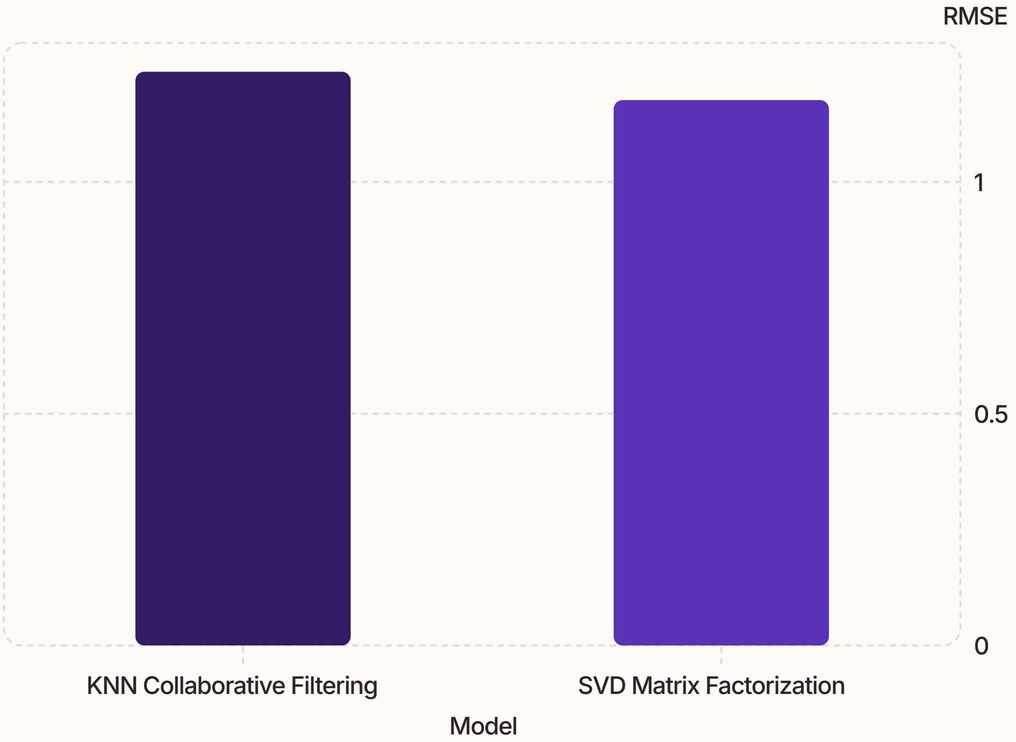
Result

RMSE = 1.1765

Pros

Handles sparsity better and yields improved accuracy.

Results and Comparison



SVD achieves the best performance (lowest RMSE).

Movie Title	Predicted Rating
The Shawshank Redemption	4.8
Inception	4.6
Interstellar	4.5

Sample personalised recommendations (predicted ratings).

Conclusions

Achievement

Built a movie recommender and performed EDA on the Netflix Prize dataset.

Model Comparison

Compared KNN item-based CF (RMSE 1.2377) and SVD (RMSE 1.1765); SVD outperformed.

Recommendation

Prefer matrix factorization for sparse academic datasets; interpretability remains